Chapter 1

Introduction

Data movement between processing and memory imposes growing energy and latency costs on modern computing. Large machine-learning models and continuous sensor streams place particularly heavy demands on that movement [1]. Section 1.1 describes the architectural origin of this limit. The approach taken here uses the dynamics of a physical device to perform part of the computation. The biological synapse motivates this approach because a single element transmits a signal, modifies its strength according to recent activity, and retains that activity over several timescales [2].

This thesis asks whether ion motion in a soft polymer host can give a two-terminal organic memristive device the fading memory needed for temporal computation. The question connects solid-state electronics, polymer physical chemistry, and computational neuroscience.

1.1 Architectural Limits of Modern Computing

General-purpose digital computers use a stored-program architecture in which processing and memory are physically separate. On contemporary hardware, moving operands between those units can cost substantially more energy than the arithmetic operation [3, 4]. Smaller transistors and faster clocks do not remove this transfer cost. The resulting limits are architectural as well as technological: the *memory wall*, the *energy wall*, and the slowing of classical transistor scaling all arise from the separation between stored data and the circuitry that acts on it [5, 6].

1.1.1 Von Neumann Architecture and Its Limitations

The architectural model of modern digital computers is the *stored-program* concept set out by von Neumann in 1945 [7]: instructions and data share one memory and are fetched word by word by a central processing unit across a connecting bus. This is what makes a single machine general-purpose, and engineering refinements (registers, caches, branch prediction, memory controllers) have left the topology essentially unchanged. The same topology is the source of the limitation. Because the processor and the memory are *physically distinct* subsystems, every instruction fetch, operand load, and result write-back must cross the channel between them; no arithmetic can begin before its operands arrive, and no result is preserved until it is moved back. This is the *processor–memory dichotomy*.

Because only a bounded amount of data can cross that channel per cycle (the *von Neumann bottleneck* [8]), performance is set by the slowest interface, not by the arithmetic. Pipelining, multi-level caches, speculation, prefetching, and out-of-order issue all mitigate this, but they preserve the

separation between computation and storage, so they shift the bottleneck without removing it. Wulf and McKee [5] named the resulting imbalance the *memory wall*: with processor speed long outpacing DRAM latency and bandwidth, an ever-larger share of each cycle is spent waiting on memory. Modern microprocessors accordingly devote much of their silicon area, and an even larger fraction of their power, to caches, prefetch logic, memory controllers, and interconnect, whose only purpose is to keep the arithmetic units supplied with data [4]. The bandwidth and energy aspects of this imbalance are summarised in fig. 1.1.

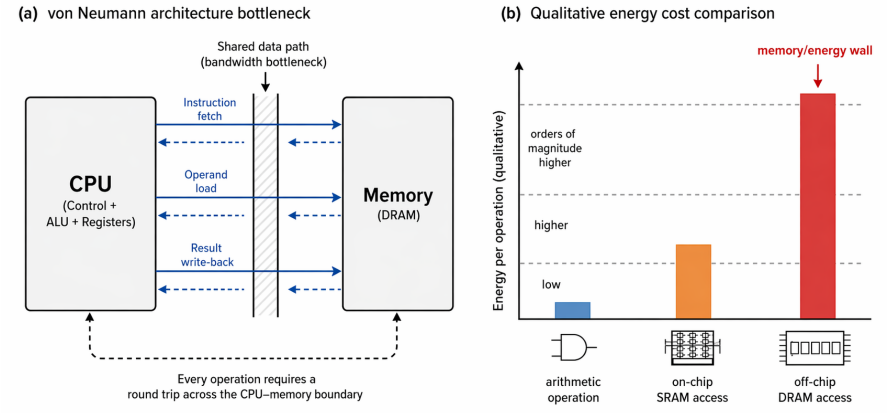


Figure 1.1: Schematic of the von Neumann architecture and the associated bandwidth and energy costs of data movement. (a) In the stored-program paradigm, the central processing unit and the memory unit are physically separated and communicate through a shared bus, so that every instruction fetch and every operand access must traverse that bus. (b) The energy cost of moving a word across this boundary exceeds, by orders of magnitude, the energy cost of the arithmetic operation performed on it, and this asymmetry has increased as technology has scaled [3].

Horowitz [3] compared the costs of elementary operations in a representative CMOS process: a 32-bit integer addition costs on the order of 0.1 pJ, an SRAM access costs tens of picojoules, and a DRAM access costs several hundred picojoules to nanojoules (table 1.1). The system-level contribution depends on how often each operation occurs. A neural-

network inference performs many multiply–accumulate operations, each requiring access to a stored weight, and training repeats these accesses across large datasets. Operand transport can therefore account for a substantial share of the energy budget. This *energy wall* arises from interconnect capacitance and from the total wire length of the memory subsystem. Spatial separation of computation and storage preserves these physical transfer costs.

Table 1.1: Approximate energy cost per operation in a representative 45 nm CMOS process, after Horowitz [3]. Memory accesses cost between one and four orders of magnitude more energy than the arithmetic they enable, and the asymmetry widens with the size of the memory addressed and with the distance the data has to travel.

Operation	**Energy**	**Class**
8-bit integer add	0.03 pJ	Arithmetic
32-bit integer add	0.1 pJ	Arithmetic
32-bit floating-point add	0.9 pJ	Arithmetic
32-bit floating-point multiply	3.7 pJ	Arithmetic
Register-file read	0.1 pJ	On-chip storage
SRAM read (8 kB)	5 pJ	On-chip storage
SRAM read (1 MB)	100 pJ	On-chip storage
Cross-chip wire (∼10 mm)	100 pJ	Interconnect
DRAM read	1.3 nJ	Off-chip storage

For several decades, Moore's increase in transistor counts [9] was accompanied by Dennard scaling [10], so successive generations provided more transistors with improved speed and energy efficiency. Dennard scaling ended around 2005, when leakage currents prevented proportional voltage scaling [6]. Transistor counts continued to rise, but performance and energy gains slowed, and power density became a limiting constraint (the *post-Dennard* or *dark-silicon* regime [11]). Process scaling alone now provides smaller reductions in the cost of the von Neumann model. At the same time, machine-learning workloads rely heavily on repeated matrix operations over memory-resident parameters [1]. Hennessy and Patterson [6] identify this period as an inflection point and argue for

domain-specific hardware that places memory and computation closer together.

1.1.2 In-Memory and Event-Driven Computing Paradigms

Two responses to this separation have emerged. The first keeps the digital workload largely intact but performs operations directly within, or very close to, the memory array; this route is described as *in-memory*, *near-memory*, or *processing-in-memory* computing. The second uses temporally structured event streams and treats the intrinsic dynamics of the physical device as part of the computation itself; this route is known as *event-driven* or *temporal* computing. The two responses, set against the von Neumann baseline they address, are summarised in fig. 1.2.

In-memory computing bypasses the classical *fetch–compute–write-back* cycle by executing operations directly inside the memory array, as in analogue matrix–vector multiplication on a resistive crossbar:

1. Each memory cell contains a conductance value that reflects a matrix entry.

2. Voltages are applied across the rows, representing the input vector.

3. The resulting column currents represent the sum of products, in accordance with Kirchhoff's laws.

This changes the array from a passive storage device into an active computing tool. Demonstrations using metal-oxide resistive memories [12], phase-change materials [17, 18], and organic ionic devices [19–21] have achieved matrix–vector operations with energy efficiencies that surpass those of traditional digital accelerators, and similar principles can be

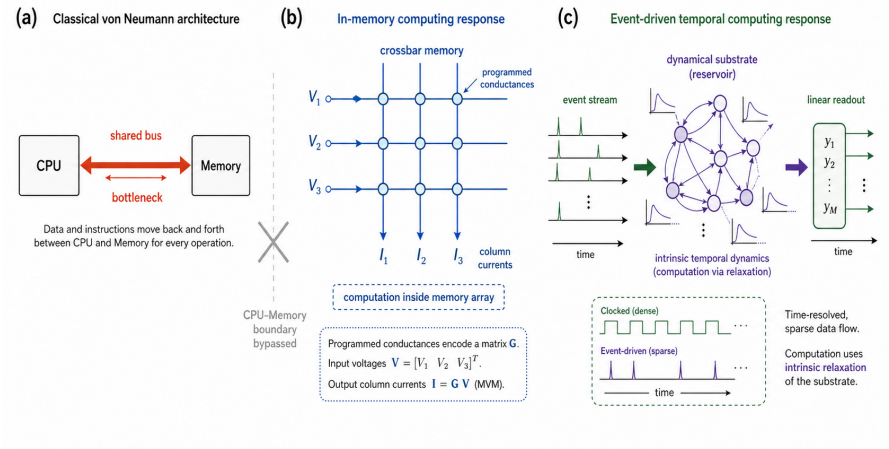


Figure 1.2: Two architectural responses to the von Neumann bottleneck. (a) In the classical stored-program architecture, every operation traverses a CPU–memory bus that fixes the bandwidth and energy ceilings of the system. (b) In the in-memory paradigm, computation is performed inside the memory array: a resistive crossbar with programmed conductances executes an analogue matrix–vector multiplication by Ohm's and Kirchhoff's laws, so the dot product no longer crosses the CPU–memory boundary [12, 13]. (c) In the event-driven paradigm, dense clocked data are replaced by a sparse stream of asynchronous events that drive a dynamical substrate whose intrinsic relaxation is itself the computation; only a linear readout layer downstream is trained [14–16].

extended to convolutional [22] and attention-like processes (the matrix-multiplication kernels at the core of transformer architectures [23]).

At the device level, however, this version of in-memory computing still requires memory properties. Each cell should hold a quasi-static conductance that encodes a trained weight, and which should be set once, read many times, and remain stable enough to meet the application's numerical precision requirements. Non-volatility over seconds to years becomes a design goal.

Event-driven computing is motivated because many signals of interest carry information in time. Natural sounds, visual transients, motor commands, biomedical waveforms, and streaming sensor outputs are all

non-static examples. Their information is distributed across amplitude, waveform evolution, and event timing, so the computational system must evolve with the input. At the algorithmic level, this is the domain of spiking neural networks [24]. At the circuit level, it corresponds to event-driven asynchronous logic. At the device level, which is the level of this thesis, it requires materials whose intrinsic relaxation dynamics are matched to the signals they process.

Biological nervous systems handle sensorimotor and cognitive tasks at substantially lower power than current artificial systems. Relevant architectural features include sparse event-based signalling, synaptic responses that depend on recent activity, and electrochemical processes that store and modify state. Neuromorphic engineering applies selected principles from these systems, including sparse-event coding [25], the use of slow device dynamics as a computing element, and functional diversity supplied by device heterogeneity [26].

Boyd and Chua [27] showed that any time-invariant nonlinear operator with *fading memory*, in an appropriate function space, can be approximated by a finite Volterra series (a power-series expansion in time-shifted inputs). This means that a system whose current state retains a decaying record of past inputs can perform various temporal computations when paired with a trainable linear readout. This concept was later formalised through algorithms such as Jaeger's *echo state network* [14, 28] and Maass, Natschläger, and Markram's *liquid state machine* [15], which were combined under the umbrella of *reservoir computing* [29, 30]. In this framework, an untrained dynamical system (the reservoir) is driven by the input stream, with only the readout being trained. When the reservoir is realised in a physical substrate, the approach is known as *physical reservoir computing* [16], with implementations in optical, mechanical, spintronic, volatile-memristive, and organic electrochemical systems [31–34].

The relevant device metrics depend on the application. Trained-weight storage requires long retention. A reservoir instead requires a finite interval over which previous inputs influence the current output: the fading-memory timescale. Controlled variation between device timescales can then give a common readout access to temporal structure over a wider range than any individual device could cover [16].

1.2 The Biological Synapse as the Engineering Template

In neural circuits, a single physical element both transmits a signal and stores part of the history that shapes future transmissions: the synapse. The same molecular event that relays information from one neuron to the next also alters its future strength, so signalling and memory share a substrate. The device target is therefore synaptic signalling and plasticity, not full neuron emulation.

1.2.1 Neural Signal Transmission: A Brief Overview

A neuron is a polarised cell specialised to receive, integrate, and transmit electrical signals [35]. Dendrites collect inputs from other cells, the soma integrates them, and when the membrane potential at the axon hillock crosses a threshold the axon propagates an outgoing signal whose terminals release a chemical messenger onto downstream cells. A mature animal neuron typically receives 10^3–10^4 such contacts [36], and each is a *synapse*. How a neuron computes is therefore governed far more by the number and strength of its synapses, and by how they change over time, than by the soma itself.

The axonal signal is the *action potential*: a stereotyped, millisecond-scale

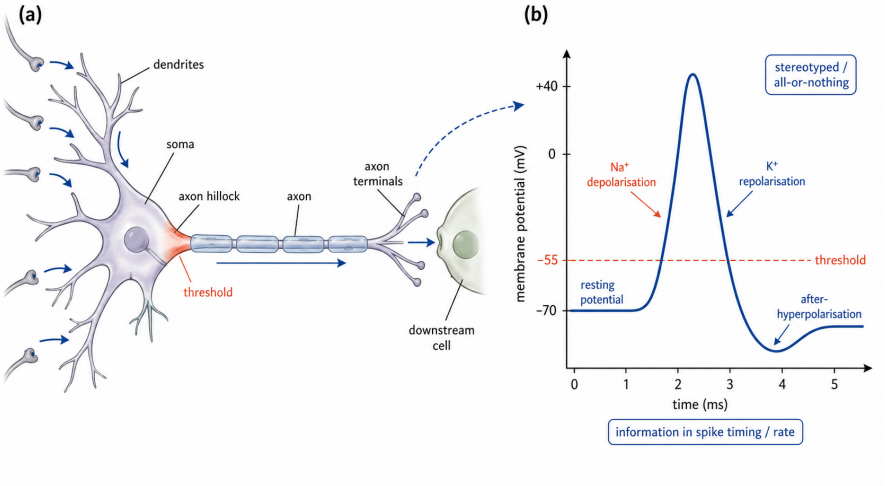


Figure 1.3: Anatomy of a neuron and the action-potential waveform. (a) A neuron collects synaptic inputs through its dendrites, integrates them at the soma, and fires an outgoing electrical event when the membrane potential at the axon hillock crosses a threshold; the resulting waveform propagates unidirectionally along the axon to the synaptic terminals, where it triggers chemical transmission onto downstream cells (the zoom shown in fig. 1.4). (b) The action potential is a stereotyped millisecond-scale depolarisation produced by the opening of voltage-gated Na^+ channels, followed by a K^+-mediated repolarisation and a brief after-hyperpolarisation before the return to the resting potential; its all-or-nothing character means that information about the input is carried by the timing and rate of these events, not their amplitude, which is the cellular basis of the event-driven view of neural computation introduced in section 1.1.2.

depolarisation that is all-or-nothing, so that information is carried not in its amplitude but in the timing, rate, and coincidence of spike trains [37, 38]. Neural computation is in this sense event-driven (section 1.1.2) and arises from the pattern of signals arriving at many synapses. The neuron's anatomy and the action potential waveform are summarised in fig. 1.3.

Synaptic transmission is largely chemical. An action potential arriving at the terminal opens voltage-gated Ca^{2+} channels, and the Ca^{2+} influx triggers neurotransmitter-filled vesicles (glutamate at excitatory cortical synapses) to fuse with the presynaptic membrane [39]. The transmitter

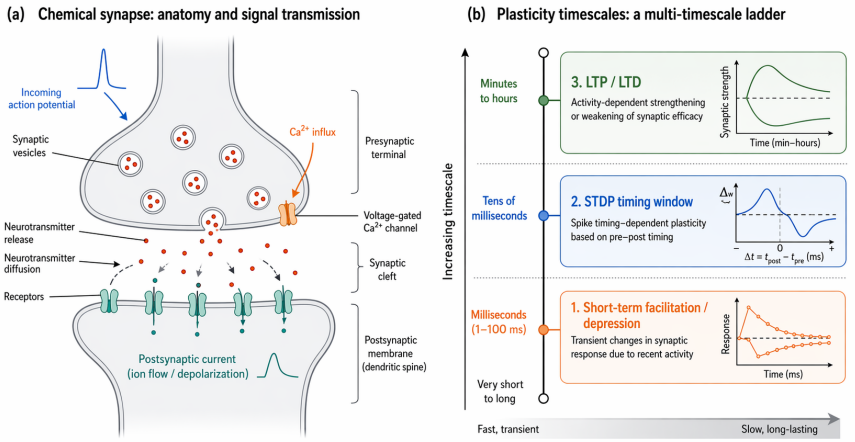


Figure 1.4: Anatomy and plasticity regimes of the chemical synapse. (a) A presynaptic terminal releases neurotransmitter vesicles into the synaptic cleft in response to a calcium-triggered action potential, and postsynaptic receptors convert the released signal into an excitatory postsynaptic current. (b) The same synapse supports a graded hierarchy of plasticity regimes (short-term facilitation and depression on millisecond timescales, spike-timing-dependent plasticity on tens of milliseconds, and long-term potentiation and depression on timescales from minutes to hours), each associated with distinct microscopic mechanisms.

diffuses across the ∼20 nm cleft and binds postsynaptic receptors [36], producing an excitatory or inhibitory postsynaptic current (EPSC or IPSC) whose amplitude and kinetics define the *strength* of the synapse at that moment. The same molecular machinery carries the signal and sets the strength of future transmissions.

The synapse combines transmission, modulation, and memory: it passes a signal, adjusts its amplitude and time course, and stores changes produced by previous activity. This co-location provides a biological reference for electronic devices that combine state and computation. The relevant anatomy and plasticity regimes are summarised in fig. 1.4.

1.2.2 Synaptic Plasticity: The Physical Basis of Learning

Experience leaves a physical trace in the nervous system, stored mainly at synapses. Hebb's 1949 postulate, that a presynaptic cell which repeatedly helps fire a postsynaptic cell strengthens its connection to it [40] ("cells that fire together, wire together"), captures two features that matter for hardware: the rule is *local* to the two cells a synapse joins, and *sensitive to correlation*. Synaptic plasticity implements Hebbian or Hebbian-like behaviour across several timescales.

On the shortest timescales, *short-term plasticity* [41, 42] makes the immediate strength of a synapse depend on its recent activity. *Facilitation* is a transient increase in EPSC amplitude when a second spike follows the first after a short delay, usually attributed to residual presynaptic Ca^{2+}; *depression* is a transient decrease caused by vesicle depletion or receptor desensitisation. With time constants from tens of milliseconds to several seconds, short-term plasticity makes the synapse a nonlinear, history-dependent temporal filter, so that identical spike sequences produce different responses depending on their timing.

On longer timescales, *long-term potentiation* (LTP) provides persistent, activity-dependent strengthening. First demonstrated by Bliss and Lømo in the rabbit hippocampus [43] and later shown to last days to weeks [44, 45], hippocampal LTP relies on NMDA receptors that open only when the postsynaptic membrane is already depolarised (a voltage-dependent Mg^{2+} block [46, 47]), making them Ca^{2+}-permeable coincidence detectors; the resulting Ca^{2+} influx drives additional AMPA receptors into the postsynaptic membrane [48, 49]. The complementary *long-term depression* (LTD) [50, 51] weakens synapses under low-frequency or anti-correlated activity, principally through reverse AMPA-receptor trafficking [52]. Together, LTP and LTD give the bidirectional, long-lasting amplitude

changes considered a primary physiological basis of learning.

Spanning these regimes, *spike-timing-dependent plasticity* (STDP), described in cortical pyramidal cells [53] and quantified in hippocampal cultures by Bi and Poo [54], makes the sign and magnitude of the change depend on the delay between pre- and postsynaptic spikes within a window of a few tens of milliseconds: a presynaptic spike before the postsynaptic one potentiates, the reverse depresses, and large delays in either direction produce no lasting change. The synapse thus holds a short memory of recent presynaptic activity, compares it with postsynaptic firing, and converts that timing relationship into a longer-lasting amplitude change.

Across these mechanisms, the synapse operates over several timescales: milliseconds for individual spikes, tens to thousands of milliseconds for short-term facilitation and depression, and minutes to hours and beyond for LTP and LTD. These partially overlapping molecular mechanisms led Abbott and Regehr [2] to describe the synapse as a computational unit. Both long-lasting and decaying state changes are relevant to hardware implementations.

1.2.3 Why Hardware Emulation of the Synapse?

The motivation for hardware synapses follows from section 1.1. In traditional computing, a software neural network retrieves data from a memory hierarchy physically separate from its arithmetic units. Each forward pass incurs data-movement costs that increase with the number of simulated synapses. A typical human brain contains approximately 10^{14}–10^{15} synapses [55] and consumes about 20 W [56]. This comparison motivates hardware that co-locates analogue state and computation; biological synapses provide one reference architecture. The order-of-magnitude envelopes of these implementation regimes are compared in fig. 1.5.

This research programme is known as *neuromorphic engineering*, whose

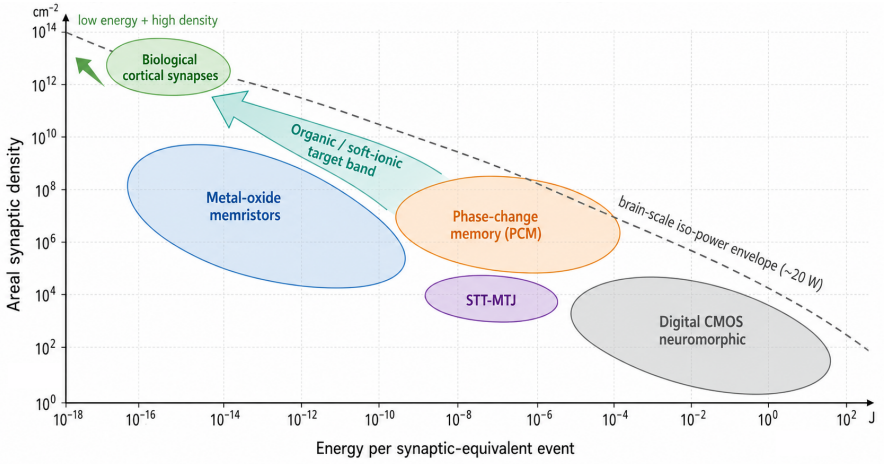


Figure 1.5: Approximate energy and density envelopes of synaptic-equivalent operations across implementation regimes. Each cluster represents a class of physical realisation: biological cortical synapses, digital CMOS neuromorphic implementations in which a synapse occupies an SRAM cell plus an arithmetic unit, the inorganic memristor families surveyed in section 1.4, and the soft-ionic composite target region forward-referenced to section 1.6. The diagonal reference line marks the brain-scale iso-power envelope ($\sim 20\,\mathrm{W}$ for $\sim 10^{14}$–10^{15} synapses) introduced at the start of section 1.2.3. Numerical bounds are conservative orders of magnitude drawn from the reviews cited in the surrounding text; individual devices may sit outside the envelopes shown.

programmatic origin is usually attributed to Mead [26]. It develops circuits in which device physics, topology, and dynamics implement selected computational characteristics of nervous systems. The synapse is a principal device-level reference.

A hardware synapse should satisfy several requirements:

- It should provide an analogue, multi-state conductance that can be set reversibly by electrical stimuli and read non-destructively with a small probe current.
- It should also be compact enough for large arrays, so that many thousands of elements can be addressed with a manageable number

of input and output lines [25, 57].

- Its response should be nonlinear in a meaningful way, for example by displaying a pulse-number or pulse-amplitude threshold below which little change is induced [13, 57].

- Finally, its dynamics should match the signals it is meant to process, whose relevant timescales span from milliseconds to seconds and beyond.

Analogue, multi-state conductance allows one device to encode several distinguishable values; a binary element would need several devices per connection to represent the same graded weight. In ionic devices, the write energy can also be low because the state is set by moving ions over short distances instead of charging large digital circuits [20, 21]. Reversibility is equally necessary. A device that can be potentiated but not reliably depressed cannot follow the bidirectional rules of section 1.2.2 and so cannot fully replicate synaptic learning [13]. These requirements define the device target used in the rest of the thesis: analogue, reversible, multi-state conductance changes set by low-energy electrical pulses.

Biological synapses combine quasi-stable memory through LTP and LTD with temporal filtering through short-term plasticity and STDP. Hardware can implement either a persistent or quasi-persistent programmable weight, as in crossbar arrays for in-memory computing [13], or a dynamic fading-memory element. This thesis focuses on the latter role through the reservoir-computing approach to event-driven temporal computing introduced in section 1.1.2 [16, 31–33].

This leads to a materials question. In biology, the timescales of short-term plasticity are determined by processes such as Ca^{2+} buffering, vesicle recycling, and receptor desensitisation, which vary across synapses [42]. A hardware implementation requires a corresponding means of adjusting

its electrical timescale. Memristive devices provide one two-terminal implementation of analogue, history-dependent behaviour.

1.3 The Memristor: Theory and History

A memristor or memristive device shows a current–voltage (I–V) response that depends on an internal state, which is influenced by the device's previous electrical activity. The ideal form is the *memristor*; the broader experimental category is the family of *memristive devices*. A relaxing internal state remains consistent with the memristive-system definition. Its decay supplies the fading-memory behaviour used later in the thesis.

1.3.1 Memristive Systems as History-Dependent Devices

A memristor is, intuitively, a resistor with memory: a two-terminal element whose resistance depends on the charge that has previously passed through it. Chua [58] introduced it in 1971 on symmetry grounds, completing the set of fundamental passive elements (resistor, capacitor, inductor) with a fourth that relates charge $q = \int i \, \mathrm{d}t$ to flux linkage $\varphi = \int v \, \mathrm{d}t$. It is described by $v = M(q)\, i$, where the *memristance* $M(q)$ has the units of resistance but, unlike an ordinary resistor, carries a record of the device's past.

If M is constant the element is an ordinary resistor; if M varies with q, the present response depends on the history of the device. Under periodic excitation this produces a *pinched hysteresis loop* in the i–v plane [59]: the loop passes through the origin (zero voltage gives zero current instantaneously) but its shape reflects the internal state left by prior stimulation. Pinched hysteresis is the standard experimental

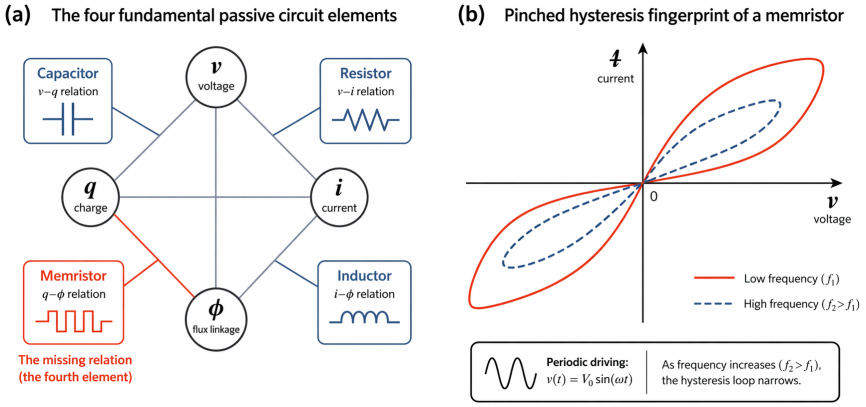


Figure 1.6: The memristor as a constitutive element and its experimental fingerprint. (a) The four fundamental two-terminal passive circuit elements, arranged by the pair of constitutive variables they relate: the resistor (v–i), the capacitor (v–q), the inductor (i–φ), and the memristor (q–φ), which completes the quadrant [58]. (b) A memristive device driven by a periodic voltage exhibits a pinched current–voltage hysteresis loop that narrows with increasing driving frequency and that passes through the origin at each zero-crossing; this loop is the canonical experimental fingerprint of memristive operation and is observed in all of the device families surveyed in the remainder of the chapter [59–61].

signature of memristive behaviour [60], and it is collected together with Chua's constitutive argument in fig. 1.6.

Few real devices satisfy the ideal relation $\varphi = \hat{\varphi}(q)$. The broader notion, introduced by Chua and Kang [61] in 1976, is the *memristive system*: an input-driven internal state $\mathbf{x}$ evolves under generally nonlinear dynamics and gates the instantaneous electrical response. This model allows $\mathbf{x}$ to include relaxation terms that drive the device back toward equilibrium once the input is removed. A volatile device whose state decays is therefore still a memristive system, even though it is not an ideal memristor. This is the framework adopted for the polymer-composite devices reported here.

1.3.2 The First Physical Realisation: TiO_2 Memristor

For more than three decades after Chua's proposal, the memristor remained mostly a theoretical construct. Thin-film metal-oxide resistive switches with hysteretic current–voltage characteristics had already been reported since the late 1960s [62–64], and many of them can now be recognised as memristive systems. At that time, however, they were not interpreted within Chua's formalism. The field still lacked a standard physical device that connected the theoretical idea to a tangible thin-film structure.

In 2008, that connection was made possible by Strukov, Snider, Stewart and Williams [65] at HP Laboratories. Their device consisted of a thin titanium dioxide (TiO_2) layer sandwiched between two platinum electrodes. Within the film, a reduced TiO_{2-x} region acted as an oxygen-deficient conductive layer, while the neighbouring TiO_2 region served as a resistive barrier. When a voltage was applied, oxygen vacancies moved, shifting the boundary between these two regions [66]. Because the total resistance depends on the relative thickness of the conductive and resistive parts, it could be modulated continuously by moving this boundary back and forth. The boundary position is the internal state, oxygen-vacancy drift updates it, and the resulting i–v response reproduces the pinched hysteresis predicted by the theory.

This result transformed the memristor from a formal circuit element into a concrete experimental object and launched a large body of work on resistive switching and memristive devices. A two-terminal resistance that can be programmed by electrical pulses, tuned over several conductance values, and read at low voltage is close to the analogue weight element required by neuromorphic hardware [67]. Crossbar demonstrations of in-memory matrix–vector multiplication followed soon after [12, 13]. The

2008 TiO_2 device is, however, only one mechanism (oxygen-vacancy drift in a transition-metal oxide), while the memristive formalism is far broader.

1.3.3 Classification of Memristive Mechanisms

In device physics, the term *memristor* describes phenomenology independently of a particular microscopic mechanism. A device is "memristive" if its current–voltage response exhibits the pinched hysteresis that Chua described and if its internal state can be both set and read electrically. This can be achieved through many different physical processes, commonly grouped into broad mechanistic families [13, 67].

These taxonomies combine two different classification axes. The first is the *spatial pattern* of the state change, whether filamentary or interface/distributed, depending on whether the conductance change is localised to a narrow conductive path or extended across the active region. The second is the *physical nature* of the state variable itself: ionic distribution, structural phase, magnetic configuration, or redox population. The two axes are orthogonal: an ionic device can also be non-filamentary, so the families considered here locate a device along both axes.

Filamentary switching. This is historically the most visible family. When a voltage is applied, a narrow conductive path forms through a previously insulating thin film, which can later be ruptured or reformed. In *conductive-bridge* random-access memories (CBRAM), the filament is built from metal atoms displaced electrochemically from an active electrode (like silver or copper) through a solid electrolyte [68, 69]. In *valence-change* memories, which are generally based on binary transition-metal oxides, the filament is instead associated with oxygen vacancies or partially reduced metal-cation states inside the oxide lattice [66].

These devices offer high ON/OFF ratios, fast switching, and non-volatile retention; their main limitation is stochasticity, since filament formation,

positioning, and rupture vary across devices and cycles [70]. They therefore work well as binary or coarse multi-level memories but are poorly suited to fine analogue representation. Panel (a) of fig. 1.7 contrasts the CBRAM and VCM variants of these devices, while panels (b)–(e) display the other four families.

Interface (non-filamentary) switching. In these devices, conductance changes over a distributed region. Some implementations modulate the height or width of a Schottky barrier by accumulating or depleting mobile ions near a metal–semiconductor interface [71]. Other implementations, especially in mixed ionic–electronic systems, drive a distributed electrochemical doping of the semiconductor [19, 20]. Because the state change is spatially extended, these devices can support a quasi-continuous range of conductance values. The ON/OFF ratio is often smaller than that of filamentary devices, but variability is usually lower and the current tends to scale more predictably with device area. This is the closest *spatial-pattern* class for several organic electrochemical devices, including the polymer-composite devices studied here, whose state variable is itself described under the ionic family below.

Phase-change memories. The active material is a chalcogenide alloy, usually $Ge_2Sb_2Te_5$ or a related composition, whose resistance depends on whether the material is in a crystalline or amorphous configuration. Joule-heating pulses are then used either to melt and quench the material into the amorphous state, or to anneal it back toward the crystalline state [17, 18, 72]. Phase-change memories provide long retention, large ON/OFF ratios, and high endurance. Their write energies are relatively large, and devices tend toward binary or few-state operation unless specifically engineered for intermediate states. They are mature storage-class memories used in analogue in-memory computing demonstrations.

Spintronic (magnetic) devices. Their resistance depends on the magnetic configuration of a tunnel junction, typically determined by the

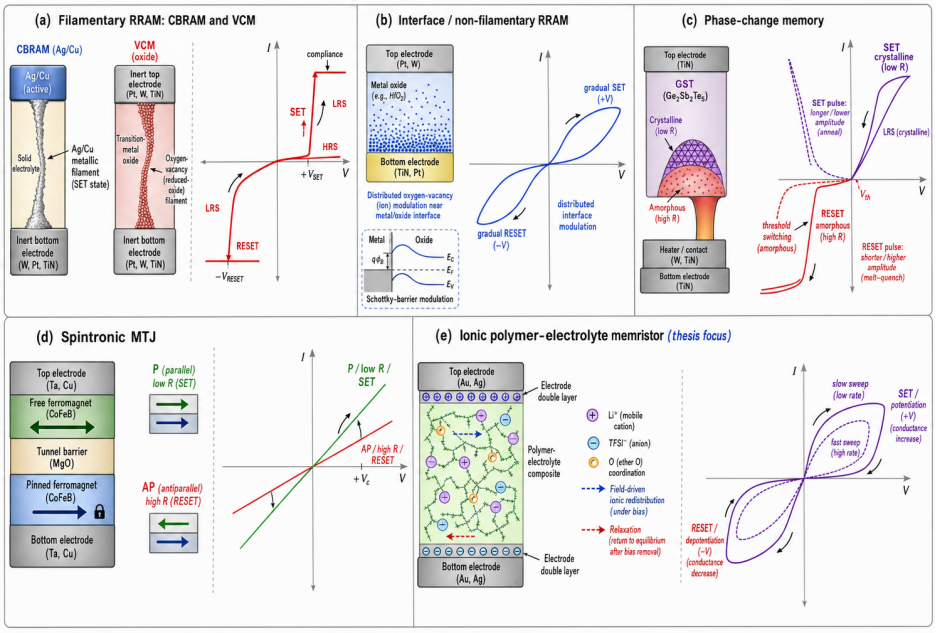


Figure 1.7: Mechanism gallery for the five memristive device families. (a) Filamentary devices, in their conductive-bridge (active-electrode metal filament) and valence-change (oxygen-vacancy filament) variants, switch abruptly between a high-resistance and a compliance-limited low-resistance branch. (b) Interface or non-filamentary devices switch through a spatially distributed change in the active layer, producing smooth, gradual, area-scaling hysteresis without an abrupt set or reset. (c) Phase-change devices toggle between amorphous and crystalline states of a chalcogenide alloy through Joule-heating melt–quench and anneal pulses, with a sharp threshold-switching event preceding the crystalline branch. (d) Spintronic magnetic tunnel junctions switch the relative orientation of two ferromagnetic layers, producing two nearly-linear branches separated by a small (order of two) ON/OFF ratio. (e) Ionic devices, in their polymer-electrolyte composite form most relevant to this thesis, redistribute mobile ions across the active layer under bias and relax back toward equilibrium when the bias is removed, producing continuous, sweep-rate-dependent hysteresis without abrupt transitions.

relative orientation of two ferromagnetic layers [73]. Spintronic devices switch very quickly at low write energy and form the basis of spin-transfer-torque magnetic random-access memory. Their limitations for analogue synaptic use are a small ON/OFF ratio (often of order two) and few stable intermediate states. Even so, their intrinsic stochastic dynamics have been put to use as a computational resource within appropriately designed architectures [74, 75].

Ionic memristors. The devices developed here belong to this family. Their internal state is the distribution of one or more mobile ionic species inside a solid matrix. The host can be an inorganic oxide, a chalcogenide, a halide perovskite, a mixed ionic–electronic conductor, or a polymer electrolyte. Likewise, the mobile species can be a small cation such as Li^+, Na^+, or Ag^+, a proton, or a small anion; in halide perovskites it is typically a migrating halide vacancy, which has been used to build energy-efficient memristive and synaptic devices [76]. The state variable is a chemical distribution of mobile species, distinguished from structural, morphological, or magnetic configurations of the host [69]. Its kinetics are governed by ionic transport, coordination, and relaxation within the matrix.

Ionic systems are intrinsically history-dependent: the distribution of mobile species at any moment reflects the electric fields previously applied. They also relax once the field is removed, as intermolecular forces and concentration gradients drive the biased distribution back toward equilibrium, so volatility is intrinsic to their operation. The associated timescales are set by ion transport and binding, both tunable chemically through the host matrix and the mobile species, the link between chemistry and relaxation that leads this thesis to the organic polymer-electrolyte subfamily.

1.3.4 From Device Metrics to Fading Memory

Memristive devices are often compared through a compact set of practical figures of merit: high- and low-resistance states, ON/OFF ratio, set/reset voltage, write energy, switching speed, retention, endurance, update linearity and symmetry, and device-to-device or cycle-to-cycle variability [13, 67, 70]. These quantities are useful only after the computational role of the device has been fixed.

For analogue in-memory computing, the device stores a *trained weight*: a conductance programmed into a crossbar cell and read repeatedly during matrix–vector multiplication [12, 13]. Long retention, high endurance, low variability, and linear symmetric updates are therefore valuable because they preserve numerical accuracy. Judged by that standard, an organic or polymer-composite device with finite retention can look weak.

For temporal computing, the same finite retention becomes the operating principle. The state of a fading-memory device should relax on a timescale matched to the input signal [15, 27]. A device used as a filter element needs a known, reproducible relaxation time; a device used as a reservoir node can also exploit nonlinearity and device-to-device differences in its temporal kernel. Figure 1.8 sketches the two modes schematically, and table 1.2 sets the corresponding parameter targets side by side.

Variability must also be separated into two kinds. Intra-device instability reduces repeatability under identical conditions. Inter-device heterogeneity is read differently by each use: it can broaden a reservoir's response by supplying different time constants and nonlinear transfer functions [16], whereas for a filter element it is the tolerance on the designed time constant, as it is for any passive component. It also complicates trained-weight arrays.

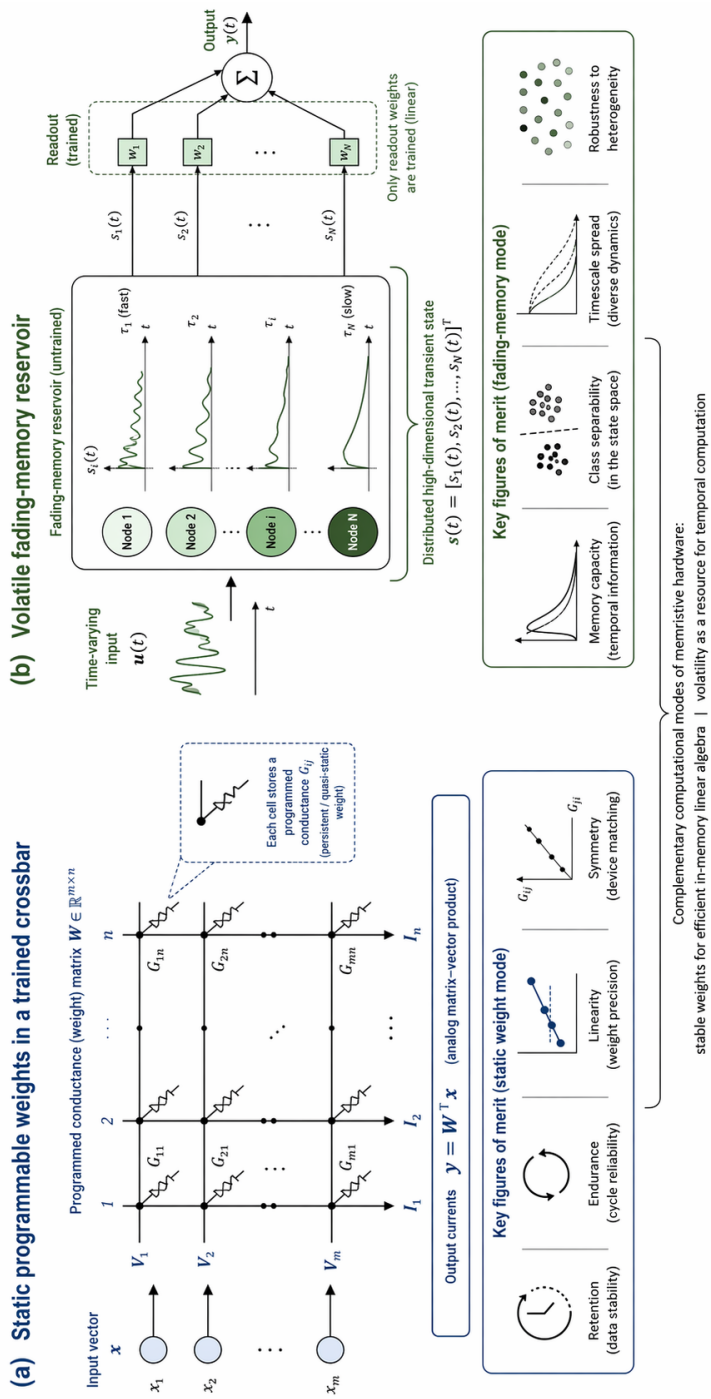


Figure 1.8: Two complementary computational roles for memristive devices. (a) A trained crossbar holds programmed conductances as non-volatile weights and performs matrix–vector multiplication by Ohm's and Kirchhoff's laws; here retention, endurance, linearity and symmetry are the relevant figures of merit. (b) A reservoir of volatile fading-memory nodes, driven by a time-varying input, projects temporal signals into a high-dimensional dynamical state; only a linear readout layer is trained, and the relevant figures of merit become memory capacity, class separability, and the spread of available time constants.

Table 1.2: Direction in which each device parameter should be optimised under the trained-weight regime (section 1.3.4) and under the fading-memory regime (section 1.3.5). For several parameters, the optimum direction reverses between regimes, so a single device cannot in general satisfy both simultaneously.

Parameter	**Trained-weight optimum**	**Fading-memory optimum**
Retention	As long as possible (s–years)	Matched to the input-signal timescale
Switching speed	As fast as possible	Matched to the input event rate
Write energy	Minimised (programmed once, read many)	Minimised within the fast-event budget
Conductance update linearity	Linear and uniform	Less critical; nonlinearity exploited as a feature
Update symmetry (± pulses)	Symmetric	Less critical; asymmetry can be exploited
Cycle-to-cycle variability	Minimised	Minimised (intra-device drift remains a defect)
Device-to-device heterogeneity	Minimised	A tolerance for filter elements; possible diversity for reservoirs [16]

1.3.5 Reservoir Computing with Volatile Memristors

Over the past decade, volatile memristive devices have been used as physical *leaky integrators*: their state rises under stimulation, decays after the stimulus is removed, and thereby carries a temporary record of recent input. Physical reservoir computing provides the computational setting for that behaviour [16].

In a volatile-memristor reservoir, the input is encoded as a stream of voltage pulses applied to an array of devices. Their internal states evolve in response to the stimuli and relax between them, so the array provides a nonlinear mapping of recent input history into a high-dimensional feature space. A linear readout trained on that space can then tackle temporal classification, regression, or prediction tasks. Demonstrations have been reported in diffusive metal-filament devices [31, 32], dynamic oxide memristors [33], and organic electrochemical systems [34].

Coincidence detection and filtering are closely related uses of the same physics. Pulses that arrive within the relaxation time of a device can accumulate, giving timing sensitivity and STDP-like responses [77]; banks of devices with different relaxation constants can process one input in parallel, with fast devices capturing sharp transients and slow devices integrating longer trends [25, 78].

A volatile device can therefore be used in two ways. In a reservoir, many devices with assorted time constants form a random feature bank, and a trained read-out selects what the task needs. In a designed circuit, one device is chosen for its relaxation time and performs a fixed operation, such as removing a slow baseline or smoothing a noisy signal; its time constant plays the role of the RC product of a passive filter. The second use matters most where long time constants are expensive. In analogue CMOS, time constants of hundreds of milliseconds already require large

capacitors or very small currents, and standard designs become bulky in silicon area [79]. Printed and flexible circuits lack compact time-dependent elements and have added temporal processing through printed low-pass filters [80]. A two-terminal film whose relaxation time is fixed by its formulation would supply seconds-scale time constants without a separate capacitor. Chapter 5 evaluates both uses from the measured device response.

Memory capacity measures how well the present reservoir state reconstructs past inputs at different delays; class separability on benchmark tasks and robustness to measured variability provide complementary tests [81]. These metrics are *task-matched*: they test whether the physical timescales of the device align with the temporal structure of the signal, which HRS, LRS, ON/OFF ratio, and retention alone do not capture.

The experimental target is therefore narrower than the full memristor literature: ionic memristive systems whose state is a chemical distribution of mobile species, whose relaxation can be tuned, and whose volatility can be used as fading memory.

1.4 Inorganic Memristors for Neuromorphic Computing

The inorganic families that dominate neuromorphic hardware are metal-oxide resistive memories, phase-change memories, and spin-transfer-torque magnetic tunnel junctions. Each is technologically important, and together they have made memristive hardware a credible post-von-Neumann substrate. Other inorganic candidates (ferroelectric tunnel junctions and FeFETs, Mott-insulator devices, and threshold-switching selectors) are narrower in analogue synaptic operation and are left outside this comparison.

1.4.1 Metal-Oxide Memristors

The most developed inorganic family is resistive switching in transition-metal-oxide thin films, in the VCM and CBRAM variants of section 1.3.3. After the TiO_2 device [65], the field expanded to HfO_2, TaO_x, NiO, and WO_x, many already familiar to silicon processing. Compatibility with back-end-of-line CMOS enabled large crossbar demonstrations [12, 68, 69]. Reported non-volatile devices reach retention of years, nanosecond switching, endurance beyond 10^9 cycles, and sub-10 nm dimensions [13, 67]. Diffusive metal-filament variants can also operate as volatile devices, with Ag or Cu clusters dissolving after programming [31–33, 82]. For analogue synaptic operation, filament stochasticity produces device-to-device and cycle-to-cycle variation in set/reset voltage and achieved conductance [70]. Nonlinear, asymmetric updates and a limited number of reliable conductance levels constrain the accuracy of in-place training on metal-oxide crossbars [13].

1.4.2 Phase-Change Memory

Phase-change memory encodes its state in the proportion of crystalline and amorphous material in a chalcogenide volume (section 1.3.3) [17]. It is a mature storage-class technology used in analogue in-memory computing. Demonstrations include mixed-precision arithmetic for solving linear systems [72] and neural-network training with PCM as the long-term weight store [83, 84]. Its thermal–structural mechanism imposes several costs: reset requires local melting, the quenched amorphous phase drifts with time and requires compensation [18], and crystallisation kinetics are tuned through alloy and geometry. These properties suit high-retention, high-endurance trained weights. They offer fewer chemical controls over fading-memory timescales than the soft-matter platform pursued here.

1.4.3 Spin-Transfer-Torque Magnetic Tunnel Junctions

A spin-transfer-torque device is the magnetic tunnel junction (MTJ) of section 1.3.3, here with a thin MgO barrier and switched between parallel and antiparallel magnetisation by a spin-polarised current through a shared read/write port [73]. STT–MTJs switch in nanoseconds at femtojoule energies, exceed 10^{12} cycles, and integrate in the CMOS back end [85]. Their intrinsic stochasticity has been used for probabilistic computing [74, 75]. For analogue synaptic operation, the parallel/antiparallel resistance ratio is small (often near two), stable intermediate states are few, and increasing their number requires more elaborate stacks [85, 86]. Retention and write energy trade off through thermal stability and switching current [85], while the crystalline ferromagnetic stack requires fabrication methods distinct from solution processing. These properties make STT–MTJs fast, non-volatile, few-state stochastic elements.

1.4.4 General Limitations of Inorganic Memristors

The three families differ in mechanism and maturity but offer limited control over the specific variables studied in this thesis. Their design parameters are primarily geometric, process-related, or stoichiometric within narrow windows [13, 70, 78]. They do not generally permit direct substitution of mobile ions with different binding chemistry or of hosts with different segmental motion. Volatile implementations exist through diffusive filaments [31–33, 82], amorphous-phase drift [18], and superparamagnetic fluctuations [74]. Their relaxation times are set by vacancy drift, crystallisation, or magnetic energy landscapes, so adjustment through a single chemical composition variable is limited. Vacuum deposition, high-temperature annealing, and top-down lithography suit dense CMOS-integrated crossbars but restrict substrate and process temperature [12,

83, 84]. Solution-processable layers allow direct reformulation and a wider range of substrates [78]. Deliberate composition changes may also separate controlled inter-device heterogeneity from intra-device instability more clearly than these established platforms do [13, 16, 18, 70].

These constraints motivate an organic ion-mediated platform that combines analogue response, chemical control of relaxation, and solution processing. Figure 1.9 summarises the qualitative comparison, while table 1.3 provides the corresponding quantitative parameters.

1.5 Organic Materials for Memristive Applications

The inorganic platforms each solve part of the problem, but none of them can be tuned by chemistry alone: reshaping the switching physics requires redesigning a rigid multilayer stack, not changing a molecule, a host, a salt, or a mixing ratio [21, 78]. Organic and polymeric materials lift that constraint. A semiconducting polymer can be tuned through backbone, side chains, and molecular weight [87]; an ion-transport host through its chelating groups, topology, and glass-transition temperature [88, 89]; and a mobile ion through charge, radius, and Lewis acidity [90]. Most such active layers are also solution-processable and processed below 150 °C, so a new formulation can be prepared by changing component concentrations in solution, and the mild processing is compatible with flexible, conformal, and biological substrates, as the stretchable artificial afferent nerve of Kim et al. [91] and the *nervetronic* systems reviewed by Park and coworkers [92] illustrate.

Organic switching can spread over many molecular sites, which makes graded analogue states more accessible than in a single-filament mechanism [19, 20]. Its relaxation timescale can also be engineered chemically.

Table 1.3: Typical operating parameters for the inorganic memristor families. Diffusive (volatile) metal-filament devices are listed separately because they are deliberately operated in the fading-memory regime, not the non-volatile one. Numerical ranges are conservative orders of magnitude drawn from the indicated reviews; individual devices may sit outside them. "cont." = quasi-continuous conductance.

Family	**Switching**	**Retention**	**ON/OFF**	**Endurance**	**Energy/write**	**Analogue states**
VCM [66, 67, 70]	1–100 ns	years	10–10^3	10^9–10^{12}	0.1–10 pJ	4–8
CBRAM [13, 67]	1–100 ns	years	10^4–10^6	10^8–10^{10}	0.1–10 pJ	2–4
PCM [17, 18, 72, 83]	10–100 ns	years (drifts)	10^2–10^3	10^8–10^{10}	10–100 pJ	8–16
STT–MTJ [74, 75]	1–10 ns	years	2–3	$> 10^{12}$	0.1–1 pJ	2
Diffusive (volatile) [31–33]	μs–ms	ms–s	10–10^3	$\sim 10^6$	0.1–10 pJ	cont. (volatile)

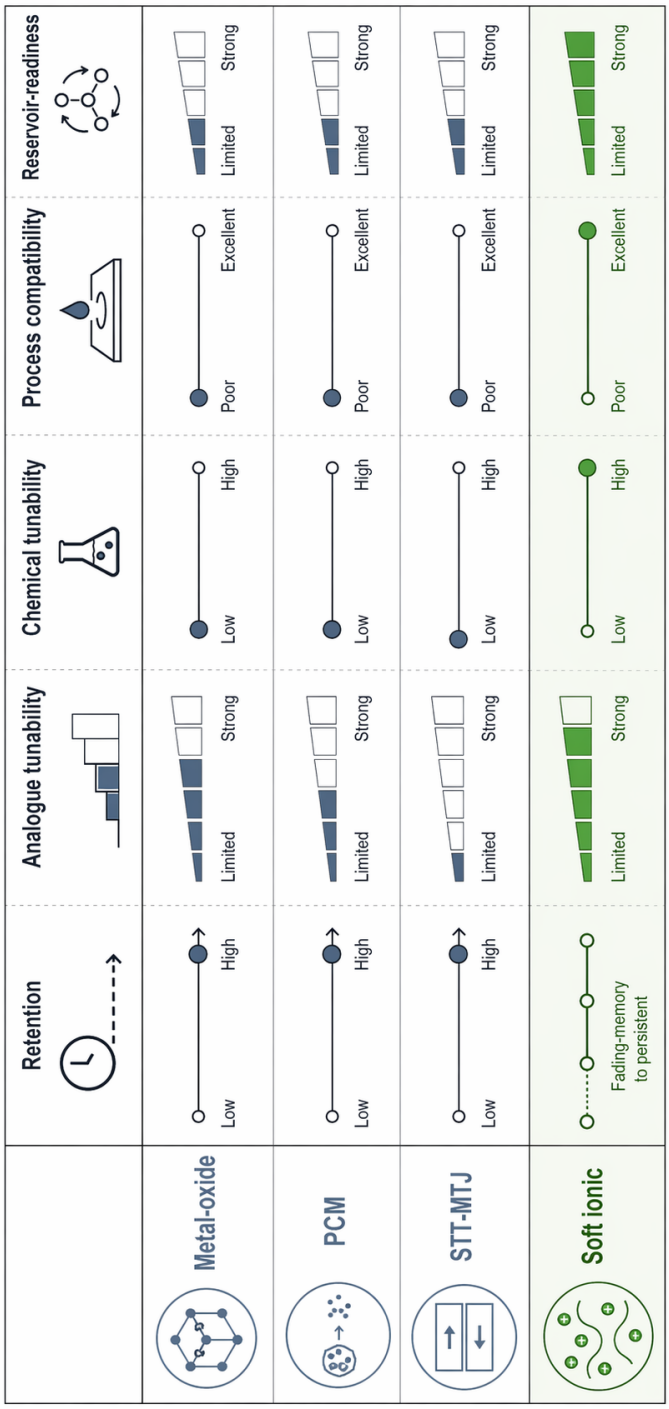


Figure 1.9: Comparison of inorganic memristor families with the soft-ionic composite route pursued in this thesis. For each family, qualitative assessments are shown along five axes: retention (the range of fading-memory to non-volatile behaviour that has been demonstrated), analogue tunability (the number of reliably programmable intermediate states), chemical tunability (the extent to which composition acts as an independent handle on the switching physics), substrate compatibility (vacuum/high-temperature workflows versus solution-processable and mild-temperature workflows), and reservoir-readiness (suitability as a fading-memory node in a temporal-computing scheme).

Ion transport in a polymer electrolyte depends on how strongly the mobile ion binds the host, on host segmental mobility, and on the number of mobile species [88, 93]. Compositional changes within one material family can therefore cover timescales from milliseconds to many seconds, the range of biological short-term plasticity and slow temporal signals [41, 42]. Cost, reproducibility, biocompatibility, and lifetime must still be established case by case. The device target is a reversible mixed ionic–electronic platform whose chemistry controls the fading-memory state.

1.5.1 Organic Mechanisms as Design Choices

As with their inorganic counterparts, the term *organic memristor* names a material family, not a single mechanism. The relevant comparison is what each mechanism (each with its own state variable, reversibility, timescale, and degree of analogue control) offers for a chemically tunable, ion-mediated, two-terminal synaptic element.

Charge trapping and detrapping. Deep or shallow traps (molecular defects, embedded nanoparticles, or boundaries between incompatible polymer phases) are filled and emptied under bias, and the trap occupation, through its local space-charge environment, sets the conductance. Trapping is common in disordered organic semiconductors and operates at modest voltages, but its analogue control is limited: the state is often binary [94], the conductance can depend non-monotonically on occupation, the trap identity is hard to establish, and the retention (set by trap depth relative to thermal energy) is not easily tuned by composition. Panel (a) of fig. 1.10 sketches this trap-occupation mechanism.

Redox switching. A redox-active species (a discrete molecule in a polymer host, a redox polymer, or a unit on a conjugated backbone) is oxidised or reduced under bias, and the distribution of oxidation states sets the conductance. The state variable is then a chemical ox-

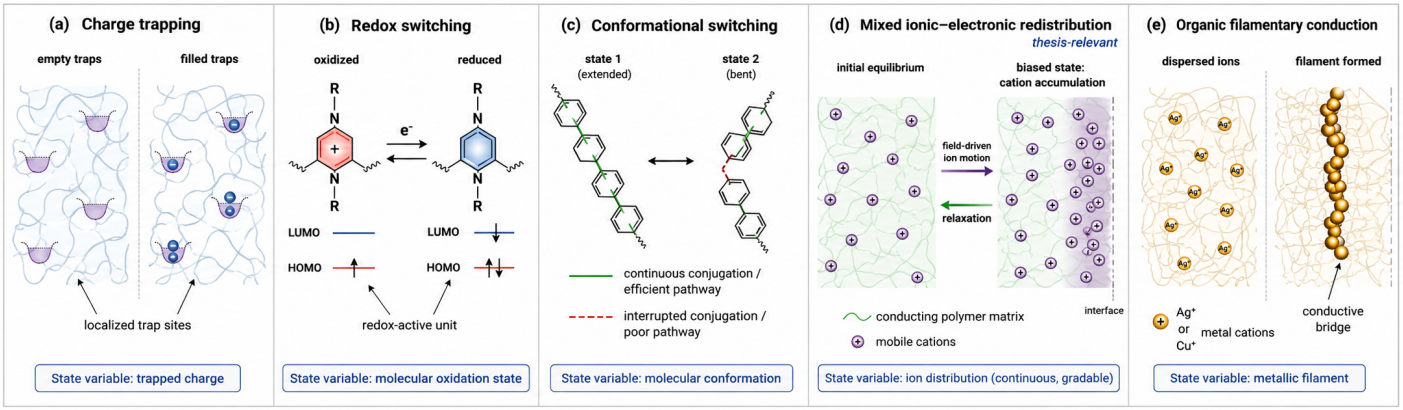


Figure 1.10: Molecular-level representation of the five mechanism families behind organic memristive behaviour. (a) Charge trapping: localised trap sites associated with defects, embedded nanoparticles, or interphase boundaries in a disordered polymer matrix, filled and emptied under bias. (b) Redox switching: a redox-active moiety alternating between its oxidised and reduced states, with the local electronic structure changing accordingly. (c) Conformational switching: a molecular unit toggling between two well-defined geometric or isomeric states with different conjugation or transport pathways. (d) Mixed ionic–electronic redistribution: a population of mobile cations within an electronically conducting polymer matrix, displaced and accumulated near an interface under bias and relaxing back toward equilibrium when the bias is removed. (e) Organic filamentary conduction: metallic cations (typically Ag^+ or Cu^+) from an active electrode migrating through the organic matrix to form a conductive bridge.

idation state with experimentally accessible potentials, reversible over many cycles in favourable cases; important examples include redox-active conjugated-polymer systems [95–97] and early electrochemical devices based on coordination complexes [98]. Not every redox process is cleanly reversible, though (side reactions, dimerisation, and slow degradation limit endurance), and purely redox devices tend to be bistable unless the response is distributed over many sites.

Conformational (structural) switching. In this case, the active layer contains a molecular species whose geometry, isomerisation state, or aggregation state can be modulated by an electric field, local heating, or coupled electrochemical processes. Examples include photochromic and electrochromic switches, as well as bistable molecular architectures whose conductance depends on molecular conformation [96, 97]. The state of these systems is related to a well-defined structural change, so they offer a high level of molecular design precision. However, the synthesis can be challenging, and the switching timescale is fixed by a particular molecular rearrangement.

Ionic redistribution in mixed ionic–electronic systems. This mechanism is the closest match to the experimental platform. The active layer combines an electronic conductor (usually a conjugated polymer) with mobile ions held in an ion-conducting host. Under bias the ions move and accumulate near interfaces, modulating electronic transport through interfacial doping, electrostatic band bending, or electrochemical doping; the conductance change is gradual, multi-state, and directly tied to the amount and arrangement of displaced ionic charge [21]. The same physics underlies light-emitting electrochemical cells (LECs) [99–105], where redistribution forms a doped junction, and organic electrochemical transistors (OECTs) [106], where a gate drives the ion exchange. In the two-terminal memristive mode the driving bias itself plays the role of the gate, and the residual ionic distribution left after the bias is removed is

what carries the memory of the stimulus.

The state variable is thus the distribution of mobile ions in the host, and relaxation after bias (governed by ion transport, the polymer binding environment, salt concentration, and morphology) sets the fading-memory process. The practical costs are that ionic motion is slower than electronic switching and that side reactions with the electrodes must be suppressed through chemistry and contact choice [107].

Organic filamentary conduction. Metallic species (typically Ag^+ or Cu^+ from an active electrode) migrate through the organic matrix and form conductive bridges similar to those found in inorganic CBRAM [68, 69]. These devices inherit the strengths of filamentary switching, such as large ON/OFF ratios and low switching voltages, but also its weaknesses, like stochastic filament formation and intra-device instability [108, 109].

The five families and their trade-offs are collected in table 1.4.

1.5.2 Organic Precedents for the Composite Strategy

The relevant precedents form a set of partial solutions to the same materials problem, not a chronological checklist: how to combine ion-mediated conductance control, analogue reversibility, and a simple device geometry. The earliest two-terminal organic memristive devices are usually taken to be the polyaniline (PANI) electrochemical junctions of Erokhin, Berzina, and co-workers [110–113], in which a PANI channel interfaced with a poly(ethylene oxide)-based electrolyte is doped electrochemically by ionic redistribution under bias. This work already paired a conjugated polymer with an ion-conducting polyether host, the same materials combination used here. It also framed the device from the start as an adaptive, analogue-tunable element supporting history-dependent conductance

Table 1.4: Mechanism families behind organic memristive behaviour.

Family	State variable	Reversibility	Analogue gradation	Timescale tunability	Refs.
Charge trapping	Trap occupation	Partial	Limited (often binary)	Low	[94]
Redox	Oxidation state of a moiety	Variable	Bistable to few-state	Low–medium	[95, 98]
Conformational	Molecular geometry / isomer	Good (per design)	Bistable / few-state	Low (per molecule)	[96, 97]
Mixed ionic–electronic	**Mobile-ion distribution**	**Good**	**Continuous**	**High (host chemistry)**	[19–21]
Filamentary (organic CBRAM)	Metallic filament	Limited (stochastic)	Few states	Low	[108, 109]

changes.

The LEC route is especially close to the platform used here. Zakhidov's *light-emitting memristor* [98], in which an ionic $[Ru(bpy)_3]^{2+}$ complex supplied both the emitter and the mobile ions, showed that redistribution in a polymer host yields memristive behaviour, while resistive switching in simple films such as poly(vinyl alcohol) [94] confirmed that the phenomenon is accessible in cheap matrices but typically binary and mechanistically opaque. The best analogue control came from the electrochemical-transistor route: Gkoupidenis' PEDOT:PSS OECTs reproduced short-term plasticity, paired-pulse facilitation, and long-term potentiation and depression [106, 114], Xu's core–sheath nanowire transistors [115] and the Van de Burgt–Fuller organic electrochemical neuromorphic device (ENODe) [19, 20] delivered linear, symmetric updates through ion exchange between electrolyte and channel [21], and halide-perovskite-gated OECTs extended the accessible timescales [116]. Each such synaptic element, however, requires a third terminal and a contacted electrolyte volume, which makes a self-contained two-terminal device an attractive complement for the reservoir paradigms of section 1.3.5.

The two-terminal line developed in parallel. Liu's biomimicking redox organic memristor [95] demonstrated coexisting short- and long-term memory in a passive two-terminal device, but also showed partial irreversibility in the redox route: the long-term regime sometimes involved permanent chemical change that limited controlled state evolution [117]. Molecular memristors, especially Goswami's metal-coordinated azo-aromatic layers [96, 97], achieved contrast, endurance, and multi-state operation through a demanding synthesis; stretchable and flexible devices [91, 92, 118, 119] demonstrated mechanical compliance; and perovskite–organic hybrids [76, 120] and organic electrochemical reservoir computing [34] broadened the field further, the latter showing that mixed ionic–electronic devices can serve as volatile physical-reservoir elements.

By the early 2020s the field had matured into a subfield consolidating around shared physical principles [21, 121], with the individually demanding features each demonstrated (ion-mediated operation in two- and three-terminal forms, analogue symmetric programming, short-to-long-term transitions, mechanical compliance, molecular precision) but not yet combined in one simple platform. The three-terminal route bought reversibility and analogue control at the cost of architectural complexity, the two-terminal redox route simplicity at the cost of full reversibility, and the molecular route precision at the cost of synthesis. In that 2022 publication context, the specific gap addressed by the proof-of-concept chapter was a two-terminal, reversible, multi-state, ion-mediated device with interpretable chemistry built from standard organic-electronic components. The route pursued here fills that gap with a composite of a conjugated semiconducting polymer, an ion-conducting polymer host, and a dissociated alkali-metal salt. These parallel developments are collected in fig. 1.11.

Contemporary physical-reservoir computing spans several node, input, output, and network architectures, so comparison requires those layers to be stated separately [122]. Since the proof-of-concept publication, organic hardware has advanced from electrochemical networks to gate- and wavelength-tuned vertical transistors with measured temporal characteristics from 0.005 s–13.2 s [123], and to reconfigurable organic electrochemical transistors that implement both reservoir and read-out functions [124]. Electrical control of relaxation is also no longer unique to multi-terminal structures: a two-terminal HfO_2 memristor has used an inter-pulse base voltage to tune its fitted relaxation time over more than two orders of magnitude [125]. No priority is claimed here for the first tunable two-terminal physical reservoir. The narrower contribution is replicated composition control of seconds-scale fading memory in a solution-processed, self-contained two-terminal organic composite, followed by model-based

rather than wired-reservoir task evaluation.

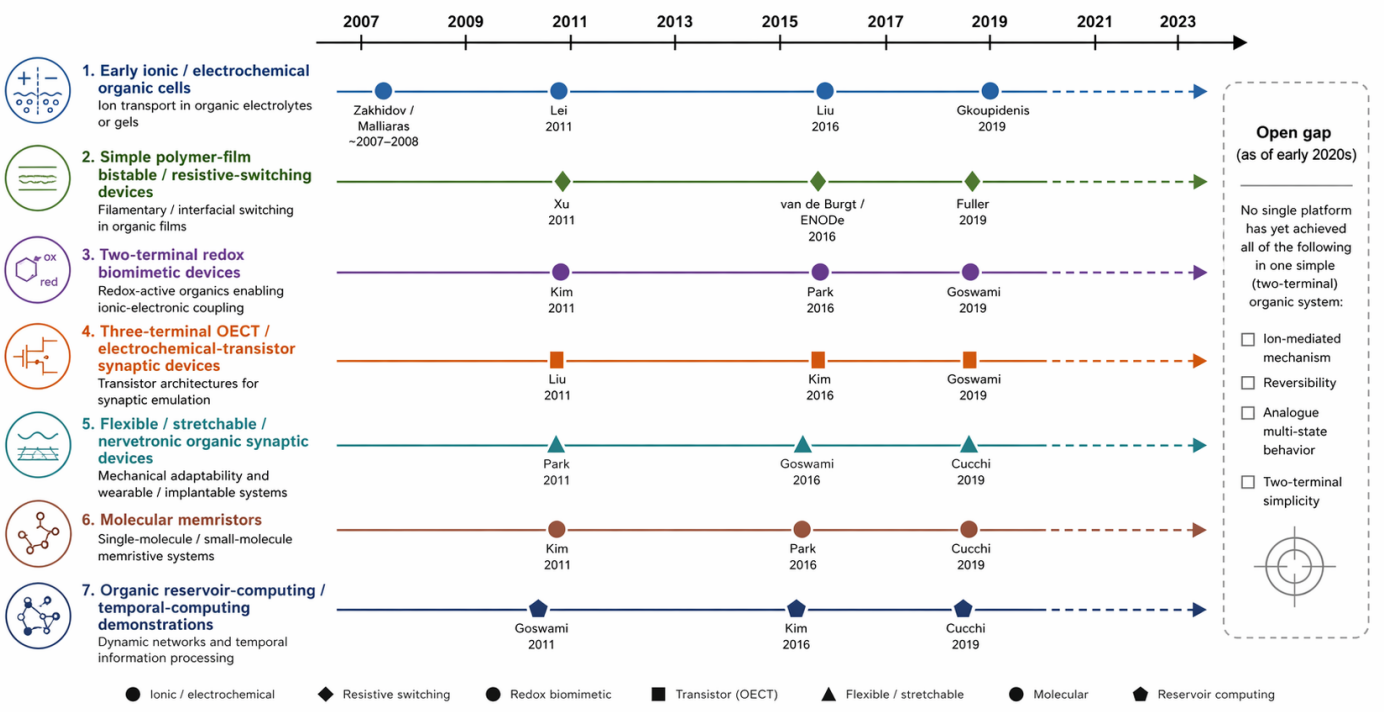


Figure 1.11: Historical map of organic memristive and organic synaptic devices, arranged by mechanism family and by year of representative publication. The map frames the literature context of the 2022 proof-of-concept publication: by the early 2020s, several demanding features of a neuromorphic organic device had each been demonstrated, but the specific combination targeted here was a self-contained two-terminal platform combining reversibility, multi-state analogue behaviour, and an interpretable ion-mediated mechanism.

1.6 Polymer Electrolyte Composites: The Enabling Strategy

The composite platform follows directly from the mixed ionic–electronic precedents above. It places a semiconducting conjugated polymer, an oxygen-rich ion-conducting polymer host, and a dissociated alkali-metal salt in one solution-processed active layer, then operates that layer in a standard two-terminal stack. Electronic transport, ion solvation, and mobile-charge supply are therefore assigned to separate components that can be changed independently.

1.6.1 Ion-Conducting Polymer Chemistry

An ion-conducting polymer is a macromolecular host in which dissolved or dispersed ions can move under an applied electric field without the help of a liquid solvent phase. Poly(ethylene oxide) (PEO) established the model system: alkali-metal salts were first shown to form solid complexes with PEO in 1973 [126], ionic conductivity was measured in those complexes in 1975 [127], and the resulting solid polymer electrolytes became practical ion conductors through the late 1970s and early 1980s [88, 89, 93]. Their relevance here is control of cation coordination, segmental motion, host crystallinity, and composition (the same variables that set memristive relaxation), not battery performance.

The polyether host has a high density of ether oxygens along the backbone or side chains. In PEO, those oxygens appear regularly along the –CH_2–CH_2–O– repeat unit, and the chain conformation in the amorphous phase allows each cation to coordinate several ether oxygens from one or more polymer chains [93, 128]. Hybrane-type hyperbranched polyester-amide hosts arrange the oxygen density differently, through ether and ester groups attached as side chains on a branched polymer core [129], but the

coordination chemistry is of the same kind.

In both hosts, oxygen donor sites can solvate alkali cations, stabilise salt dissociation, and define local coordination environments. Charge-delocalised anions such as triflate ($CF_3SO_3^-$) or bis(trifluoromethanesulfonyl)imide (TFSI, $N(SO_2CF_3)_2^-$) are commonly treated as weakly coordinating counter-ions [130]. Neither description fixes the relative macroscopic mobility: ion pairing, host segmental motion, concentration, morphology, and field also contribute.

Ion transport in these hosts is usually described through three related microscopic phenomena:

- From the *vehicular* side, a cation and its solvation shell move together through the polymer matrix, so long-range transport is coupled to polymer motion [131].

- From the *hopping* or site-to-site side, the cation instead transfers between coordination sites through a local rearrangement of oxygens, without macroscopic motion of the host. This last view is conceptually close to Grotthuss-type transport in proton conductors [132], even though the chemistry of the transfer itself is different.

- Finally, from the *segmental-motion-assisted* side, short polymer segments reorganise around the cation, creating and destroying coordination sites that allow effective diffusion [133, 134].

These three phenomena are not mutually exclusive, although the weight of each contribution during transport depends on host chemistry, cation identity, salt concentration, and temperature [88, 93].

The macroscopic ionic mobility of a polymer electrolyte generally correlates with the segmental mobility of the host, which is set by the glass transition temperature T_g and by whether the host remains amorphous

during operation. Below T_g, segmental motion is essentially frozen and ionic conduction drops. Above T_g, it is activated, and ionic transport becomes measurable, with the temperature dependence in this regime typically following the Vogel-Fulcher-Tammann form characteristic of fragile glass-formers [88, 135].

Even above T_g, however, not all of the host contributes equally. Crystalline regions contribute little to conduction and can even trap ions in ordered environments [128, 134]; the textbook picture is that conduction proceeds essentially through the amorphous fraction, although it has since been shown that specific crystalline stoichiometric phases can themselves support measurable ionic motion [136]. Low-crystallinity or fully amorphous hosts are therefore preferred for ion transport. A low T_g and a low crystalline fraction can be pursued through copolymerisation, branching, or plasticisation; hyperbranched architectures such as the Hybrane family are attractive here because branching itself suppresses crystallinity [137]. For room-temperature memristive operation the host should remain amorphous and mobile under ambient conditions, ideally without added plasticisers.

The strength of cation binding depends on cation identity, which hard–soft acid–base (HSAB) reasoning [90] makes explicit. Ether and ester oxygens in polyether and polyester-amide hosts are hard Lewis bases: their donor pairs are localised, their polarisability is low, and their coordination to alkali cations is primarily electrostatic. Among the alkali cations, Li^+ is the hardest Lewis acid (small, with a high charge density), and should therefore interact most strongly with hard oxygen donors. Na^+ and K^+ are progressively softer, with larger radii, lower charge densities, and weaker electrostatic coupling to the same oxygen environment.

In a common polyether host and at fixed composition, the residence time of the cation at a coordination site might be expected to follow the coordination-strength order $Li^+ > Na^+ > K^+$. This provides a hypothesis

for the devices' fading-memory timescale: harder cations should relax more slowly and softer cations more quickly. The experimental chapters test this hypothesis. In the archive used here, composition is the replicated quantitative variable, while cation order is not robust across host and anion. The coordination geometry, the three transport mechanisms, and the cation–oxygen binding hierarchy that motivate the hypothesis are summarised in fig. 1.12.

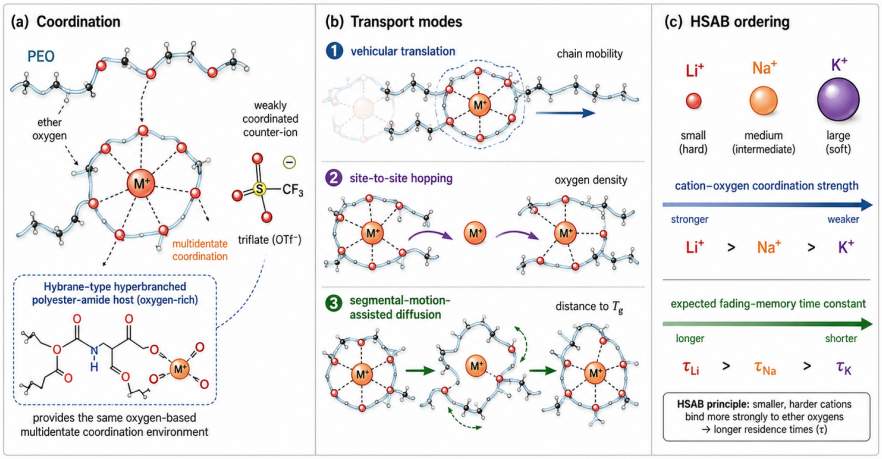


Figure 1.12: Molecular-level chemistry of ion transport in the polyether-hosted polymer-electrolyte composite. (a) A short poly(ethylene oxide) segment coordinates an alkali cation through four to six ether oxygens, with triflate shown nearby as a weakly coordinating counter-ion [93, 128, 130]; oxygen coordination in the Hybrane host of chapter 2 is a qualitative analogy, not a measured coordination structure. (b) Long-range ionic transport can combine vehicular translation, site-to-site hopping, and segmental-motion-assisted diffusion [131, 133, 134]. (c) Within a common polyether host, hard–soft acid–base reasoning motivates a $Li^+ > Na^+ > K^+$ cation–oxygen coordination-strength hypothesis [90]. Coordination strength alone does not determine mobility, and the thesis does not assume this ordering as a host- and anion-independent law.

A second set of design parameters is provided through composition. A polymer electrolyte is defined by the host, ion identity, and salt-to-host ratio; in a composite device, the polymer-to-semiconductor ratio also

contributes. The salt-to-host ratio sets the number and spatial density of occupied coordination sites, while the host-to-semiconductor ratio sets the volume available for ionic transport and the connectivity of the electronic phase. The two ratios are therefore treated as independent composition axes in the later chapters.

These considerations motivate the host choices made in this thesis. Hybrane DEO750 8500 is a hyperbranched polyester-amide with a high density of ether and ester oxygens. It provides an amorphous, low-crystallinity ion-transport host and is used in chapter 2 as the host of the proof-of-concept Super Yellow (SY)/Hybrane/lithium triflate (LiOTf, $LiCF_3SO_3$) composite. The synaptic responses were measured on assay-specific junctions sharing this formulation and architecture; they do not form a device-population study.

PEO is the host used for the broader comparative work. Like Hybrane, its ether oxygens coordinate alkali-metal cations, segmental motion enables transport, and composition can be varied while the electronic semiconductor is held fixed. PEO also has an extensive solid-polymer-electrolyte literature covering its phase diagrams, coordination geometries, conduction mechanisms, and salt-dependent dynamics [88, 93, 127, 128, 134]. The replicated composition series therefore uses a host whose structural and transport behaviour is independently constrained. A hyperbranched polyether, trimethylolpropane ethoxylate (TMPE), provides a comparison between linear and branched polyether architectures. Host, anion, and cation substitutions are surveyed at smaller sample sizes and treated as illustrative contrasts rather than material laws.

1.6.2 LEC-Derived Composite Operation

Polymer light-emitting electrochemical cells (LECs) provide an established example of stable, reversible coupling between ionic redistribution in a

polymer host and electronic transport in a conjugated semiconductor [138]. An LEC is a simple two-terminal sandwich [99]: a conjugated light-emitting polymer is mixed from a common solution with an ion-conducting polymer and a dissociated salt. Under bias, cations migrate toward the cathode and anions toward the anode, forming doped regions that inject carriers into the undoped bulk, and the resulting dynamic p-i-n profile produces electroluminescence in the intrinsic region between the doped edges [102].

Two models were historically used to describe this operation. In the electrochemical-doping model [99, 139], the dominant effect is bulk doping of the conjugated polymer; in the electrodynamic model [100, 101], it is the lowering of charge-injection barriers by ion accumulation at the contacts. Later experiments [102, 104] and a unifying framework [103] showed these to be limiting cases of the same ion redistribution under different injection regimes, not contradictory mechanisms. The two models, their reconciliation, and the bias-off configuration that the memristive composite reuses are compared on a common cathode–anode position axis in fig. 1.13; studies of electrochemical side reactions [107] additionally identify the degradation pathways that must be suppressed.

LECs operate over useful windows with lifetimes of hundreds to thousands of hours [105, 140], provided the contacts and operating protocol support repeated charge injection and withdrawal without rapid destructive side reactions [141–143]. Their electrolyte supplies mobile ions whose redistribution modulates the conjugated polymer. Sustained bias in an LEC maintains a p-i-n profile for emission; in a memristive device, a pulse instead writes an ionic distribution whose bias-off residue provides the stored state. Ionic redistribution can therefore serve as the internal state variable of a two-terminal memory device.

The most direct precedent is the Hybrane-based composite that underlies the proof-of-concept device of chapter 2. Mardegan and co-workers [105]

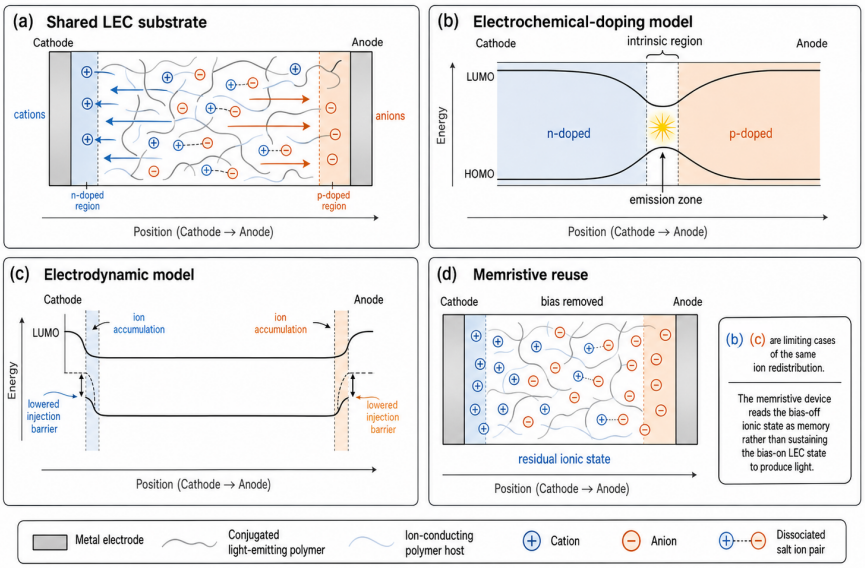


Figure 1.13: Light-emitting electrochemical cell operating models and their memristive reuse, drawn on a common cathode–anode position axis. (a) Shared substrate: under sustained bias, cations drift toward the cathode and anions toward the anode, forming doped regions at the two contacts. (b) Electrochemical-doping model: the dominant effect of the ionic redistribution is *bulk doping* of the conjugated polymer, producing a spatially wide p-i-n profile with emission localised at a narrow intrinsic region. (c) Electrodynamic model: the dominant effect of the same redistribution is *injection-barrier lowering* at the metal/polymer interfaces, producing thin ion-accumulation layers at the contacts and an essentially flat-band bulk. The two pictures were later reconciled as limiting cases of the same redistribution under different injection regimes. (d) Memristive reuse: with the bias removed, the residual ionic distribution left behind by the previous bias is the *stored internal state* of the device.

demonstrated stable LECs based on Hybrane hyperbranched polyesteramide hosts blended with a conjugated light-emitting polymer, reporting long operational lifetimes under continuous bias. Their work validates the semiconducting-polymer / ion-conducting-host / salt architecture, the solvent system, and the processing protocol that chapter 2 then reuses under pulsed bias to operate the same composite as a memristive device.

1.6.3 The Ionic Response in Time and Frequency

The ionic redistribution that stores the device state has a characteristic relaxation time, set by the host segmental mobility discussed in section 1.6.1. This timescale has two equivalent descriptions. In the time domain, a written ionic profile decays toward equilibrium over a period that the proof-of-concept and comparative chapters measure in seconds (chapters 2 and 4). In the frequency domain, it appears as a cutoff. At low drive frequencies, ionic redistribution produces the pinched hysteresis of fig. 1.6. Above the redistribution rate, the junction approaches a passive dielectric response and the hysteresis loop narrows [60, 61]. For a slow polymer-electrolyte host, this crossover falls in the sub-kilohertz range and overlaps the frequency bands of peripheral physiological signals examined in chapter 5.

The standard small-signal probe of this frequency response is electrochemical impedance spectroscopy (EIS). A small alternating voltage is applied across a range of frequencies and the complex impedance recorded; plotted as a Nyquist or Bode spectrum, it separates the bulk resistance to ion transport through the host from the capacitance of the interfacial layers that accumulate at the contacts. An equivalent-circuit model assigns these features to discrete circuit elements: for polymer light-emitting electrochemical cells and the related composites used here, the circuit of Munar and co-workers resolves a bulk ionic resistance together with geometric and chemical (double-layer) capacitances [144]. That bulk ionic resistance is the equilibrium counterpart of the driven relaxation (a more ion-rich, more mobile matrix has both a lower ionic resistance and a faster fading memory), so impedance spectroscopy gives an independent, small-signal test of the link between composition and timescale that large-signal switching alone cannot supply [88, 130]. Chapter 4 uses both probes, equilibrium impedance spectra and biased frequency sweeps, as microscopic corroboration of the composition control established from

the driven dynamics.

1.6.4 Rationale for the Composite Approach

The composite separates electronic transport, ion solvation and transport, and mobile-charge supply across three components [99]: a conjugated polymer, an oxygen-rich host, and a salt. Salt, cation, and blend ratio can therefore be changed independently of the rest of the stack [21]. The salt-to-semiconductor ratio changes the nominal supply of potentially mobile ions. HSAB reasoning motivates stronger cation–oxygen binding for harder alkali cations, while charge-delocalised triflate is weakly coordinating [130]; it does not, on its own, rank their drift mobilities. A fast/slow species asymmetry is therefore one candidate origin for multiple memory timescales [110, 112, 113]. Its effect must be tested within each host, anion, electrode, concentration, and drive protocol.

Composition is a second control variable. The salt-to-semiconductor mass ratio sets the relative supply of mobile ions and hence the achievable size of the conductance change [88, 89], while the host-to-semiconductor mass ratio sets the relative amount and connectivity of the segmentally mobile phase in which they rearrange; in principle these vary the amplitude and the rate of the response separately, though the comparative chapter treats their independence as an experimental question. The three components share a common solvent, so changing the formulation requires only different solution ratios on the same electrode geometry. This makes a replicated composition grid practical, while host, anion, and cation substitutions remain lower-power chemical contrasts.

Devices with the same polymers but different blend ratios should exhibit a spread of ionic relaxation times, with sample-limited chemistry and device-to-device variation adding further diversity. Each formulation would then define an element with its own time constant, selected at

fabrication in the way a component value is selected from a catalogue. Together, such elements form a bank of temporal kernels with distinct decay constants [16, 81]. This spread complicates precise trained-weight storage but can cover a range of fading-memory timescales [31–34]. The application chapter evaluates both the individual elements and the banks from the measured device behaviour.

The experimental chapters test this rationale first with the Hybrane/LiOTf proof-of-concept device of chapter 2, and then with the broader PEO-based corpus in which composition is varied as the replicated axis and chemistry is surveyed as illustrative side evidence. Figure 1.14 summarises this concept used throughout the thesis, linking the two-terminal stack to the qualitative timescale ladder originally expected from cation and composition chemistry.

From Background to Device

The resulting device target is a solution-processed composite of a conjugated semiconducting polymer, an oxygen-rich ion-conducting host, and a dissociated alkali-metal salt. It is operated as a two-terminal element whose residual ionic distribution forms the internal state. Chapter 2 introduces the proof-of-concept SY/Hybrane/LiOTf device, its memristive and synaptic response, and the ion-transport mechanism proposed to explain its two retention regimes.

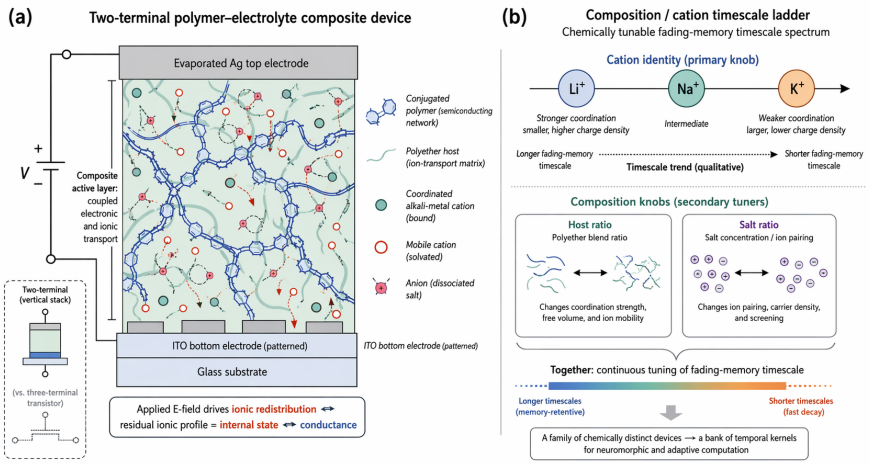


Figure 1.14: The thesis-specific polymer-electrolyte composite device and its chemically tunable dynamical timescale. (a) Vertical two-terminal stack: a solution-processed composite active layer combines a conjugated semiconducting polymer network with an oxygen-rich ion-conducting host and alkali-metal salt, between a patterned ITO bottom electrode and an evaporated silver top electrode; the working model assigns memory to a residual ionic distribution after bias. (b) Timescale ladder *as anticipated at the design stage*, with cation identity drawn as the primary control and the host-to-semiconductor and salt-to-semiconductor mass ratios as secondary ones, placing the fading-memory time along the $Li^+ > Na^+ > K^+$ coordination order. The panel records the starting hypothesis, not the result. Chapter 4 inverts this hierarchy: composition (the PEO/SY and salt/SY mass ratios) is the replicated quantitative control of the timescale, whereas the cation is sample-limited and its anticipated ordering is not borne out (the triflate retention series runs in reverse).

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